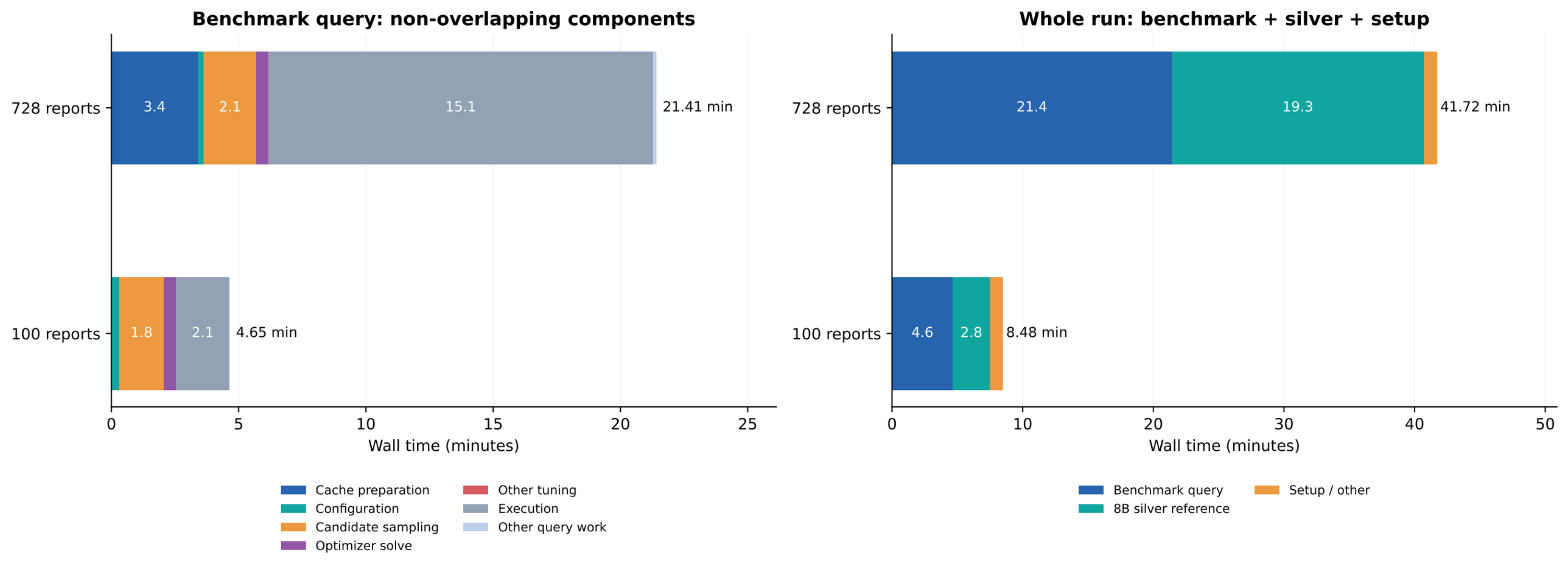
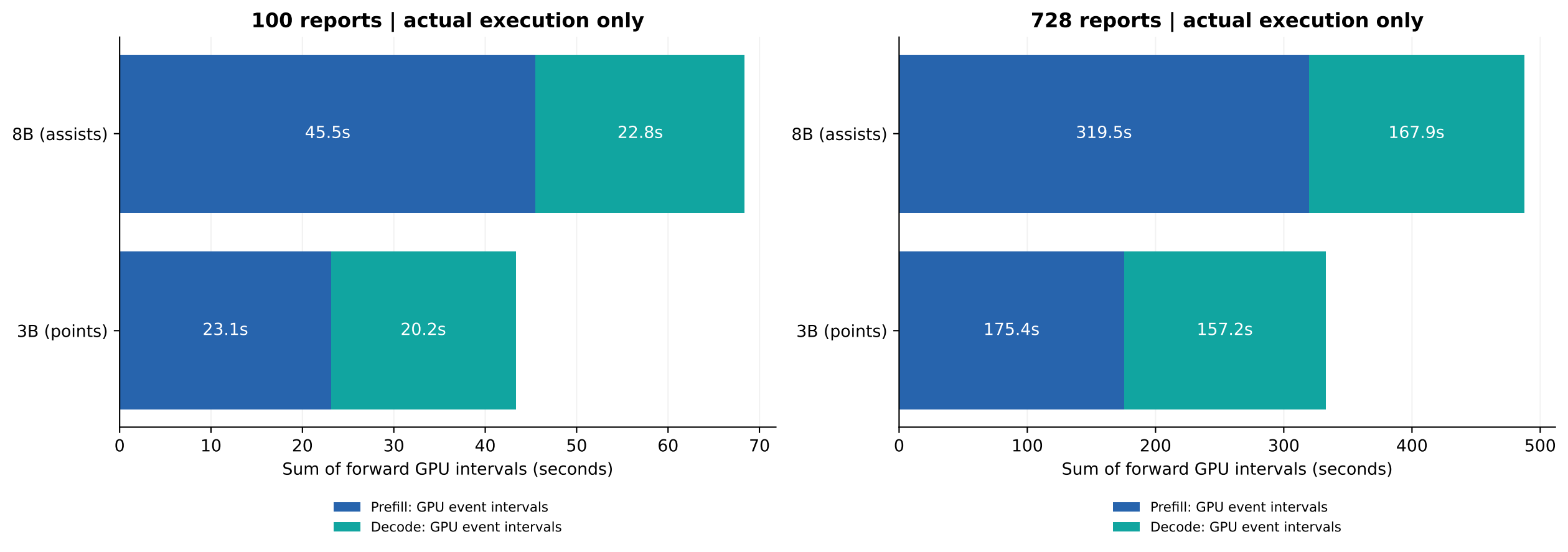
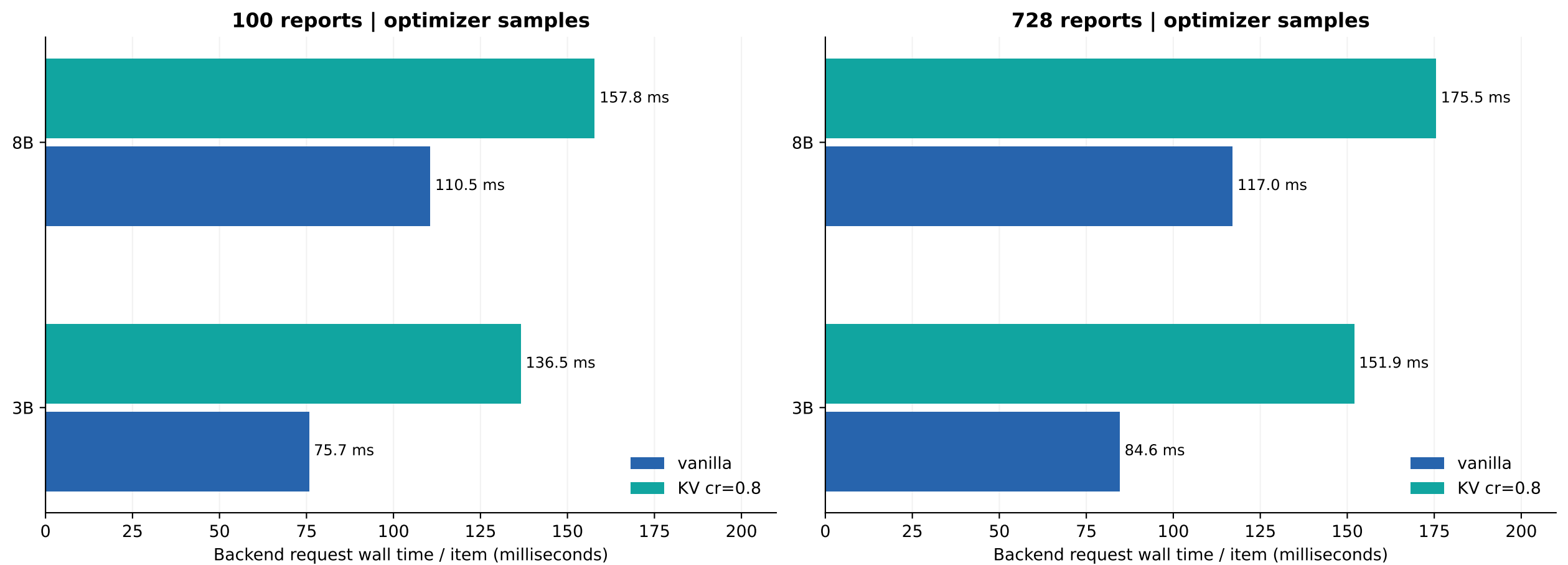
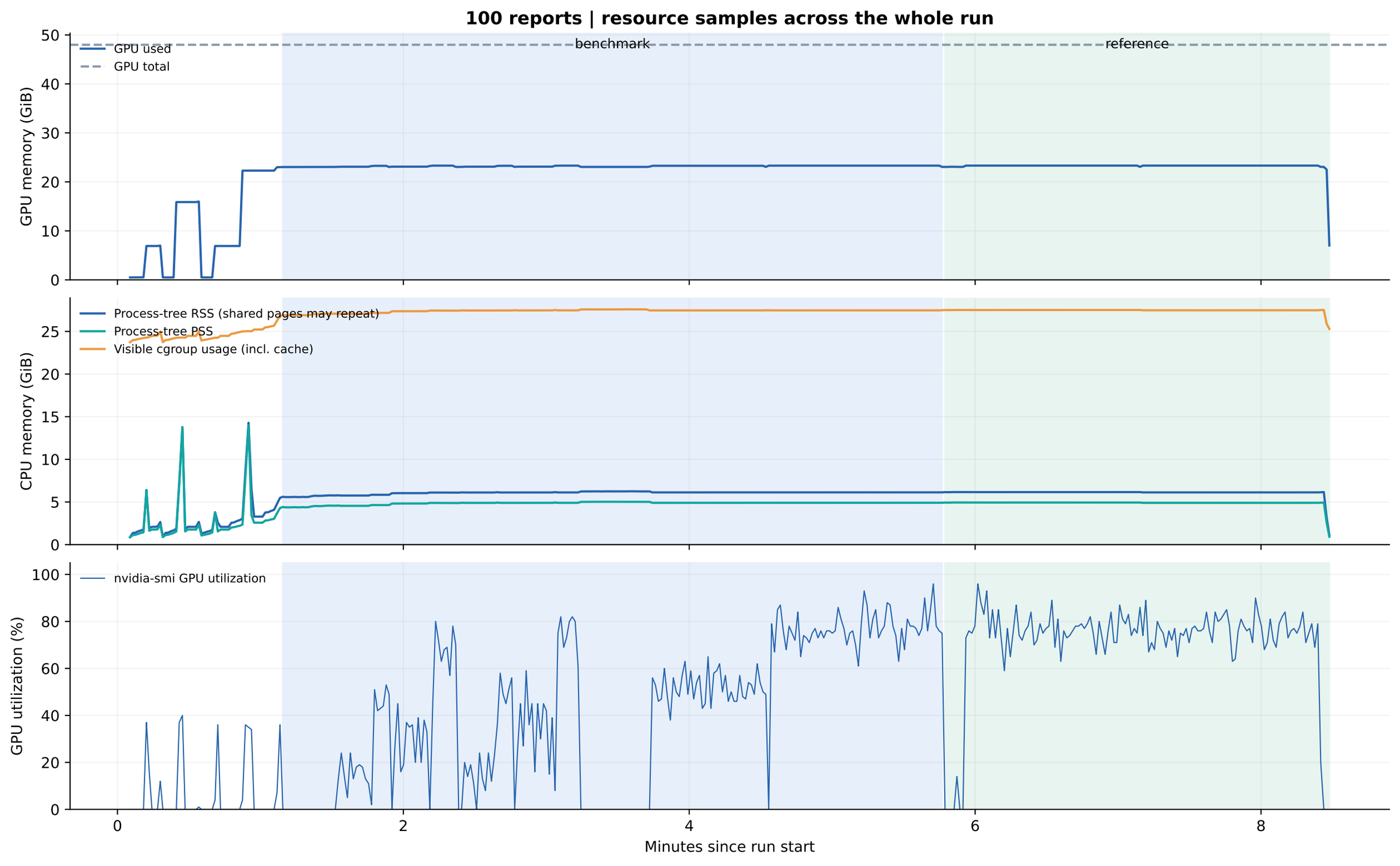
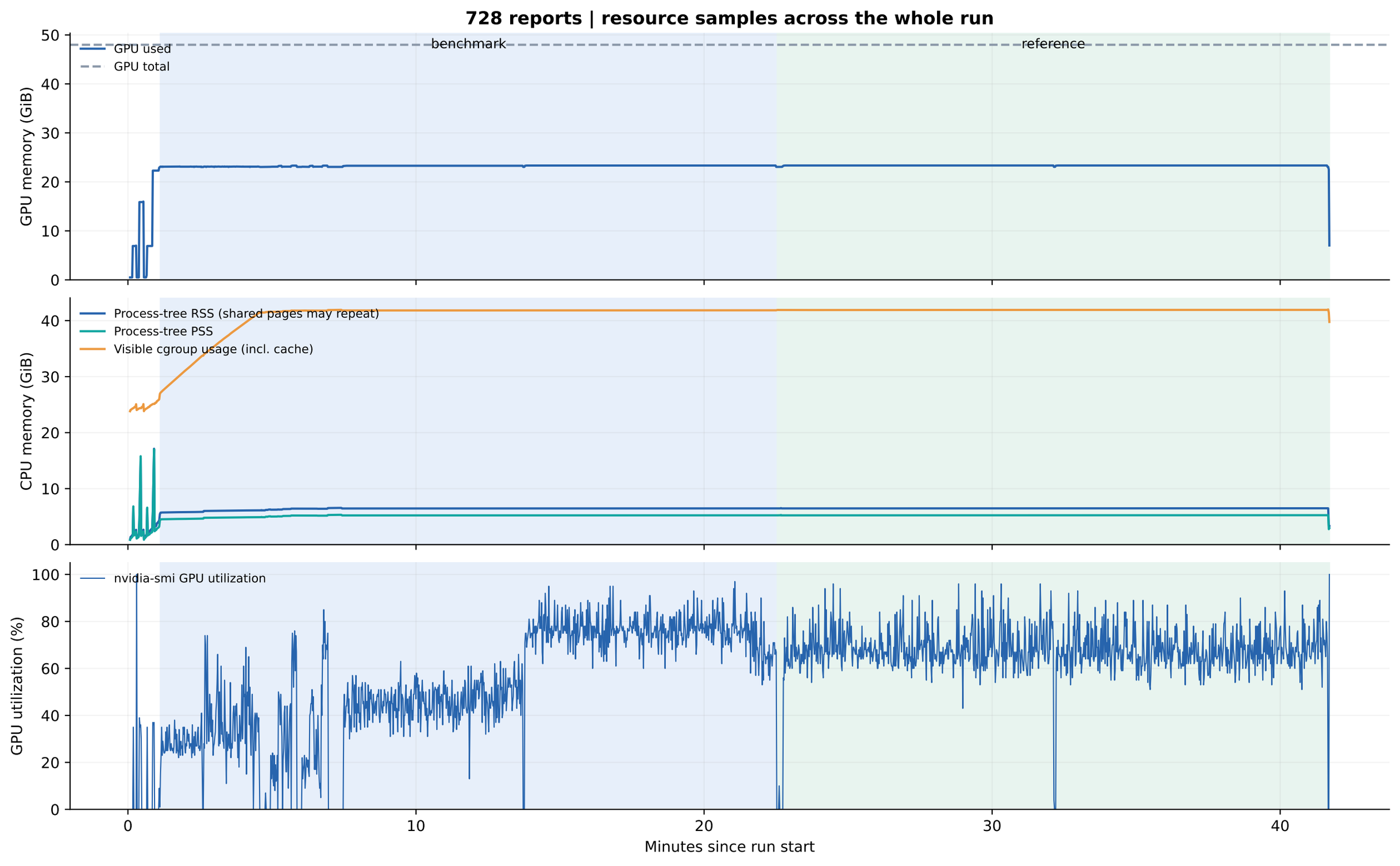
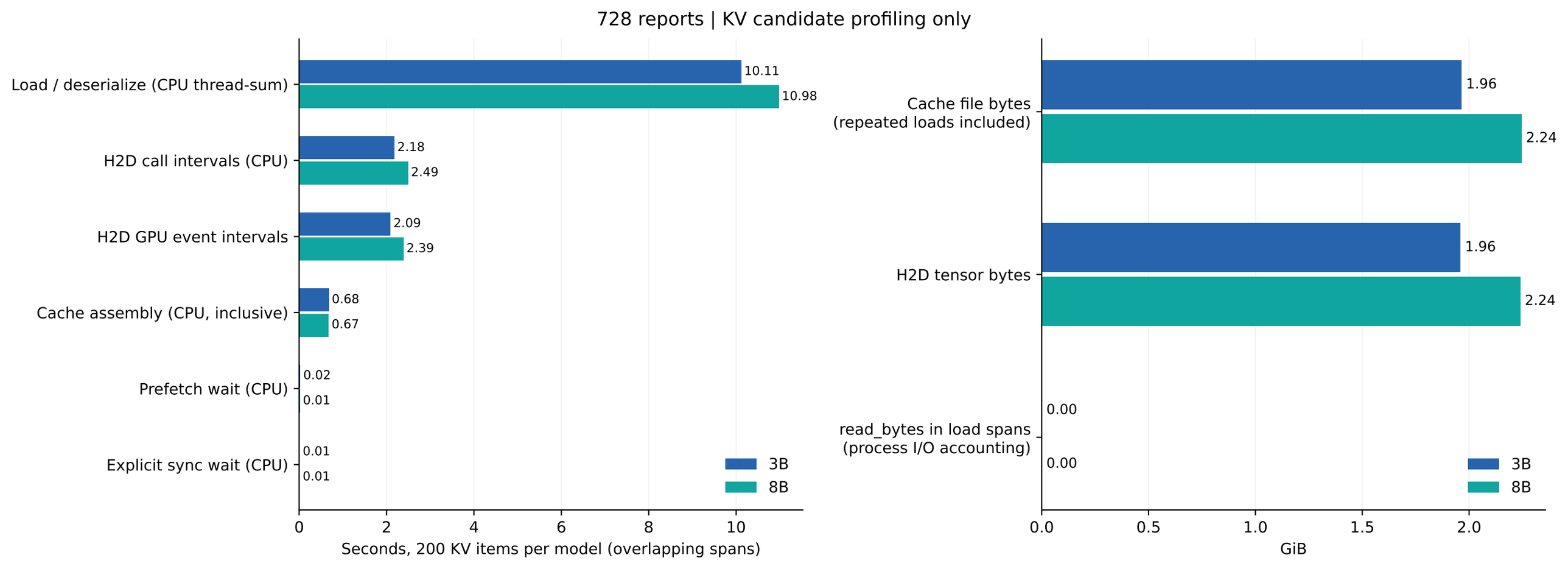

Benchmark query: non-overlapping components
728 reports
100 reports
3.4
2.1
15.1
21.41 min
1.8
2.1
4.65 min
0
5
10
15
20
25
Wall time (minutes)
Cache preparation
Configuration
Candidate sampling
Optimizer solve
Other tuning
Execution
Other query work
Whole run: benchmark + silver + setup
728 reports
100 reports
21.4
19.3
41.72 min
4.6
2.8
8.48 min
0
10
20
30
40
50
Wall time (minutes)
Benchmark query
8B silver reference
Setup / other

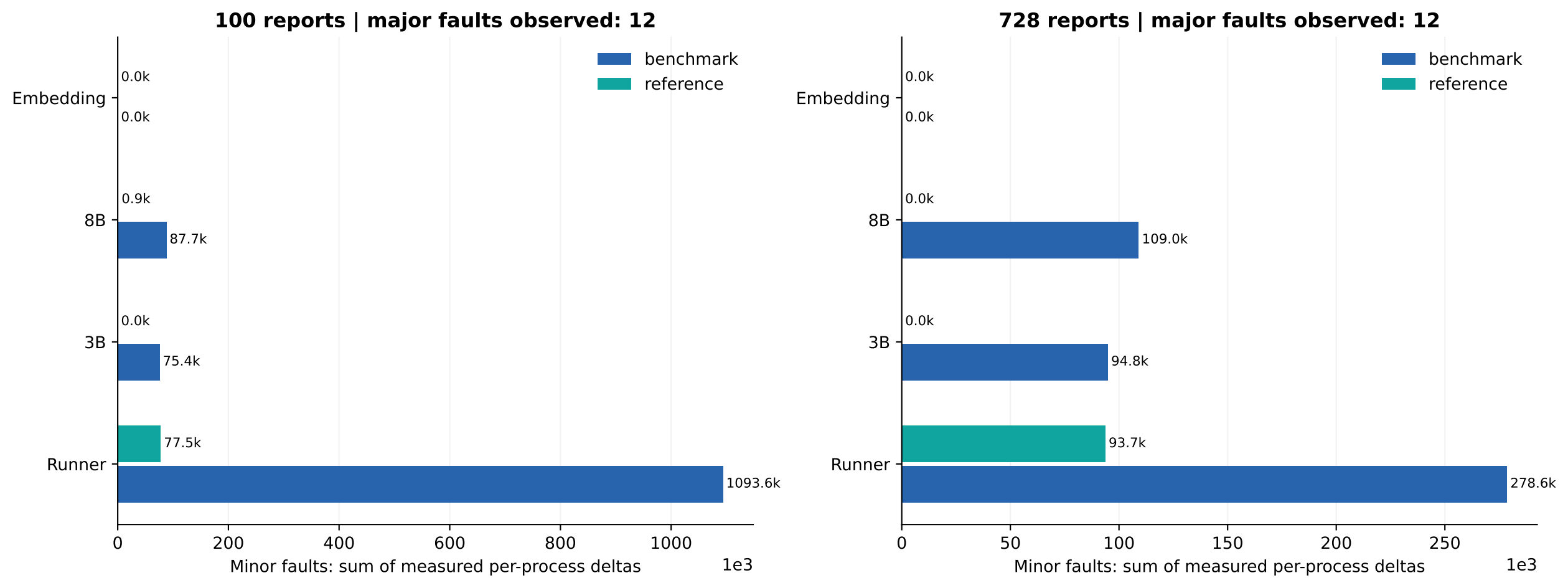

100 reports | actual execution only
8B (assists)
3B (points)
45.5s
22.8s
23.1s
20.2s
0
10
20
30
40
50
60
70
Sum of forward GPU intervals (seconds)
Prefill: GPU event intervals
Decode: GPU event intervals
728 reports | actual execution only
8B (assists)
3B (points)
319.5s
167.9s
175.4s
157.2s
0
100
200
300
400
500
Sum of forward GPU intervals (seconds)
Prefill: GPU event intervals
Decode: GPU event intervals

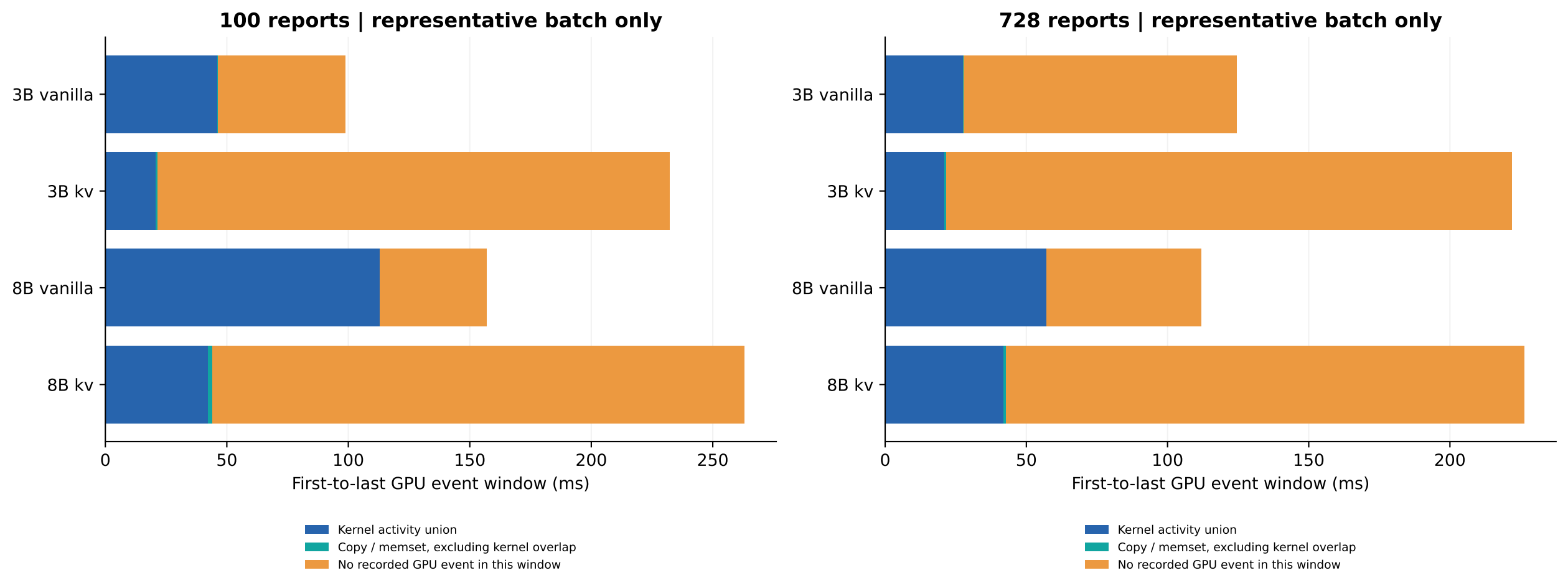

100 reports | optimizer samples
8B
157.8 ms
110.5 ms
3B
136.5 ms
75.7 ms
vanilla
KV cr=0.8
0
25
50
75
100
125
150
175
200
Backend request wall time / item (milliseconds)
728 reports | optimizer samples
8B
175.5 ms
117.0 ms
3B
151.9 ms
84.6 ms
vanilla
KV cr=0.8
0
25
50
75
100
125
150
175
200
Backend request wall time / item (milliseconds)

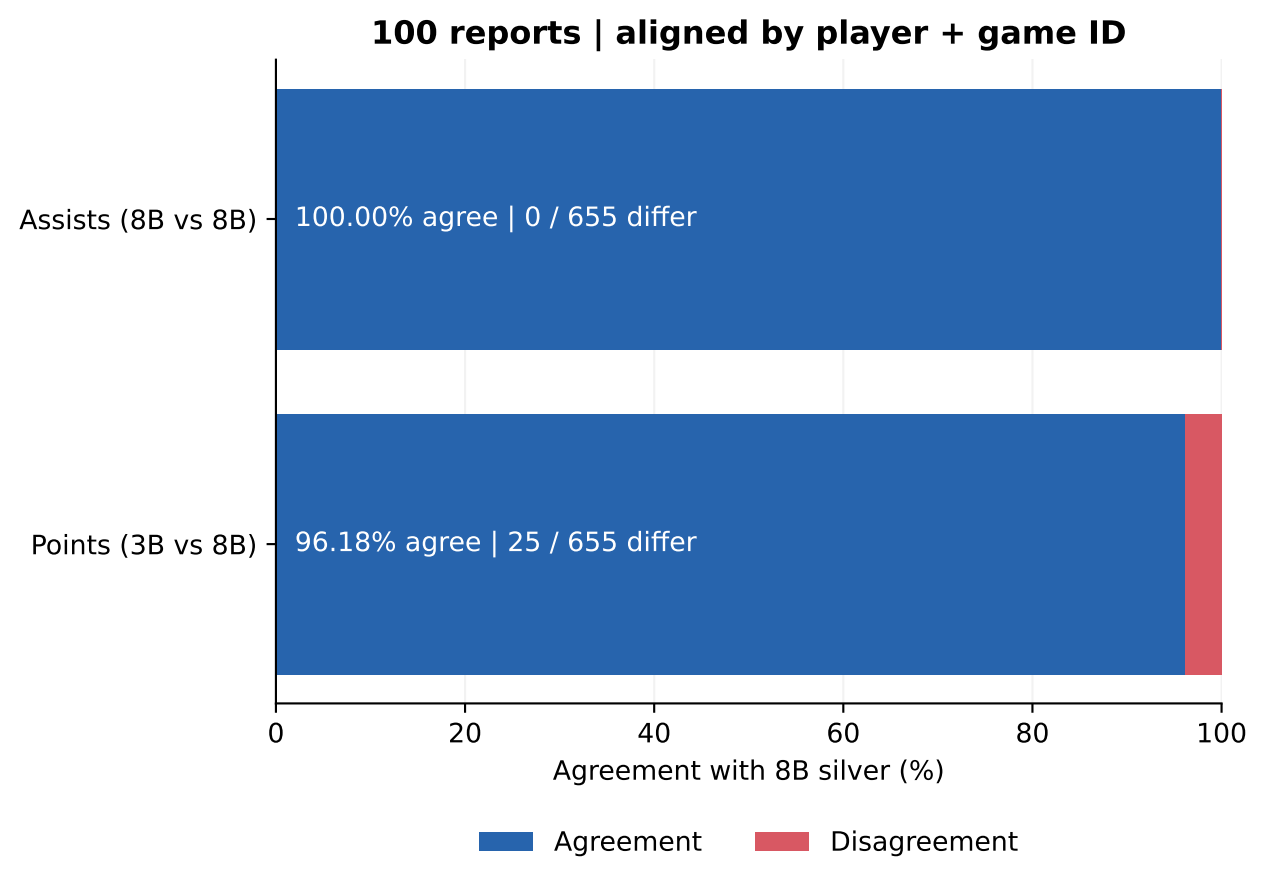

100 reports | resource samples across the whole run
benchmark
reference
GPU used
GPU total
GPU memory (GiB)
Process-tree RSS (shared pages may repeat)
Process-tree PSS
Visible cgroup usage (incl. cache)
CPU memory (GiB)
nvidia-smi GPU utilization
GPU utilization (%)
Minutes since run start

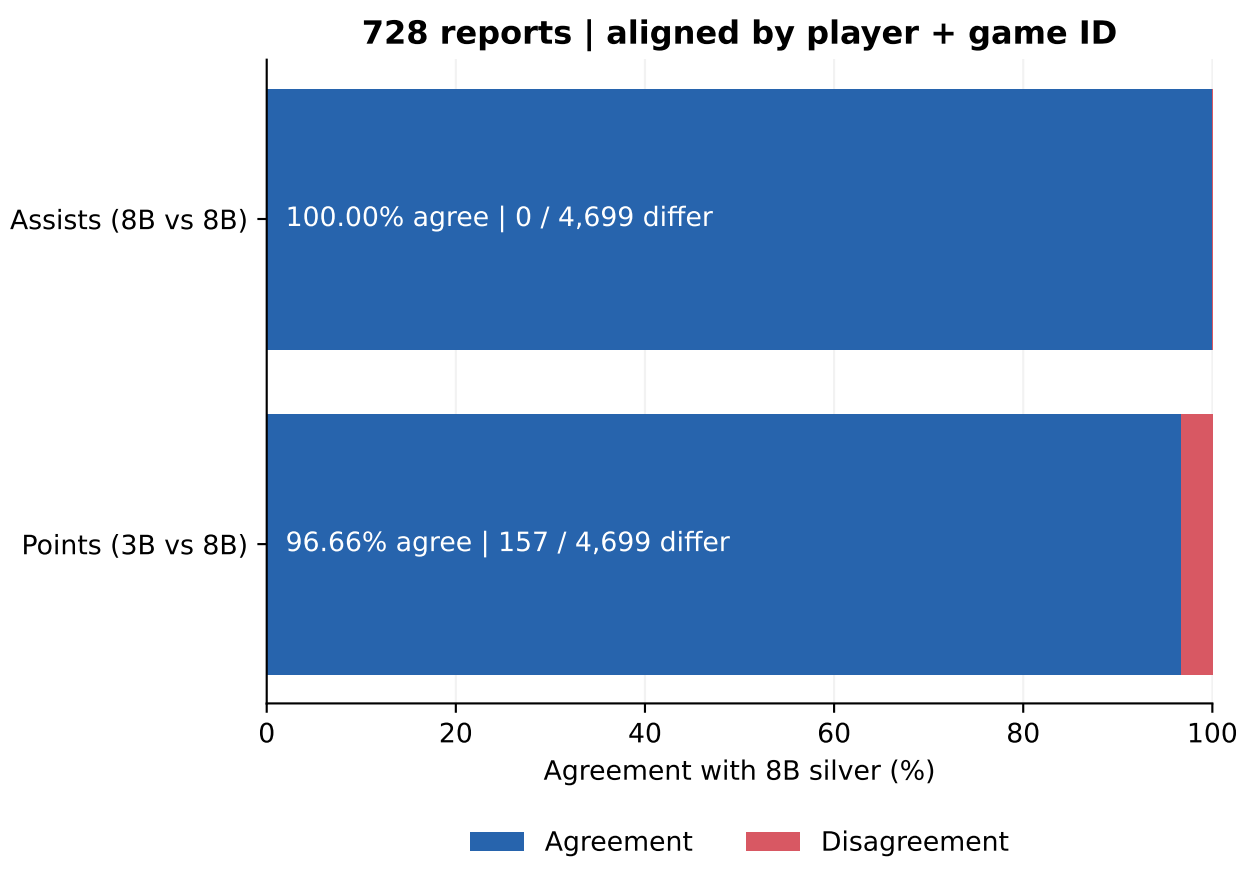

728 reports | resource samples across the whole run
benchmark
reference
GPU used
GPU total
GPU memory (GiB)
Process-tree RSS (shared pages may repeat)
Process-tree PSS
Visible cgroup usage (incl. cache)
CPU memory (GiB)
nvidia-smi GPU utilization
GPU utilization (%)
Minutes since run start

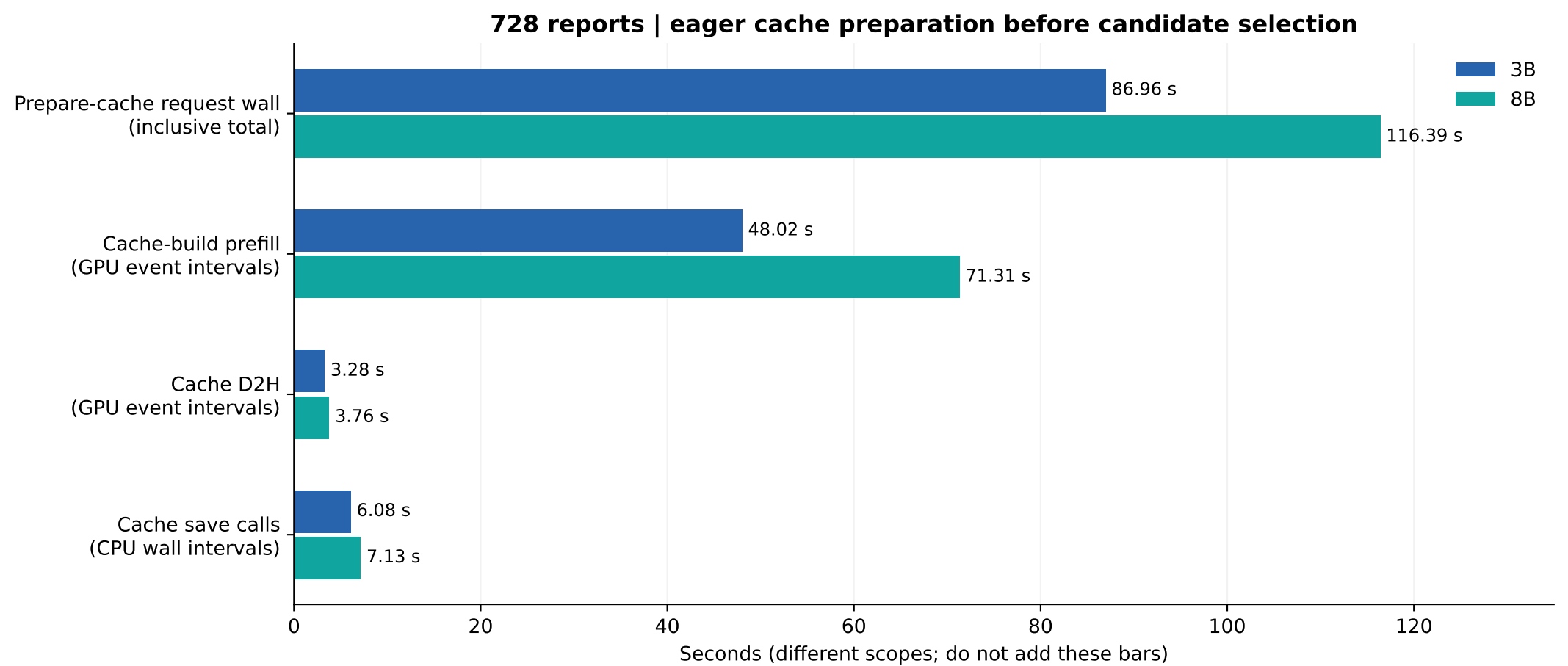

728 reports | KV candidate profiling only
Load / deserialize (CPU thread-sum)
H2D call intervals (CPU)
H2D GPU event intervals
Cache assembly (CPU, inclusive)
Prefetch wait (CPU)
Explicit sync wait (CPU)
10.11
10.98
2.18
2.49
2.09
2.39
0.68
0.67
0.02
0.01
0.01
0.01
3B
8B
0
2
4
6
8
10
Seconds, 200 KV items per model (overlapping spans)
Cache file bytes
(repeated loads included)
H2D tensor bytes
read_bytes in load spans
(process I/O accounting)
1.96
2.24
1.96
2.24
0.00
0.00
3B
8B
0.0
0.5
1.0
1.5
2.0
GiB


100 reports | major faults observed: 12
benchmark
reference
Embedding
8B
3B
Runner
0.0k
0.0k
0.9k
87.7k
0.0k
75.4k
77.5k
1093.6k
0
200
400
600
800
1000
Minor faults: sum of measured per-process deltas
1e3
728 reports | major faults observed: 12
benchmark
reference
Embedding
8B
3B
Runner
0.0k
0.0k
0.0k
109.0k
0.0k
94.8k
93.7k
278.6k
0
50
100
150
200
250
Minor faults: sum of measured per-process deltas
1e3


100 reports | representative batch only
3B vanilla
3B kv
8B vanilla
8B kv
0
50
100
150
200
250
First-to-last GPU event window (ms)
Kernel activity union
Copy / memset, excluding kernel overlap
No recorded GPU event in this window
728 reports | representative batch only
3B vanilla
3B kv
8B vanilla
8B kv
0
50
100
150
200
First-to-last GPU event window (ms)
Kernel activity union
Copy / memset, excluding kernel overlap
No recorded GPU event in this window


100 reports | aligned by player + game ID
Assists (8B vs 8B)
100.00% agree | 0 / 655 differ
Points (3B vs 8B)
96.18% agree | 25 / 655 differ
0
20
40
60
80
100
Agreement with 8B silver (%)
Agreement
Disagreement


728 reports | aligned by player + game ID
Assists (8B vs 8B)
100.00% agree | 0 / 4,699 differ
Points (3B vs 8B)
96.66% agree | 157 / 4,699 differ
0
20
40
60
80
100
Agreement with 8B silver (%)
Agreement
Disagreement


728 reports | eager cache preparation before candidate selection
3B
8B
Prepare-cache request wall
(inclusive total)
86.96 s
116.39 s
Cache-build prefill
(GPU event intervals)
48.02 s
71.31 s
Cache D2H
(GPU event intervals)
3.28 s
3.76 s
Cache save calls
(CPU wall intervals)
6.08 s
7.13 s
0
20
40
60
80
100
120
Seconds (different scopes; do not add these bars)